Computer Network
↳ main use is to share data

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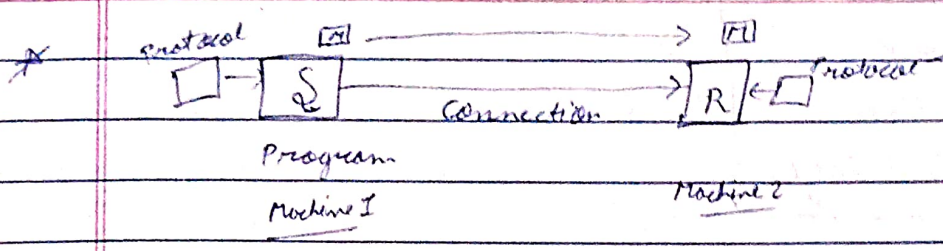
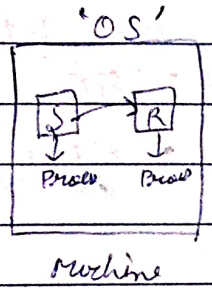
1) Physical



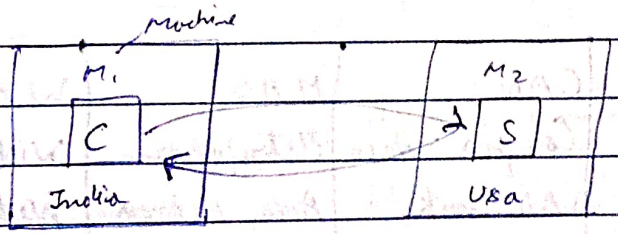
set of instructions or Rules
↳ so that receiver decode the data

Connection with protocols is Communication.



eg - Keyboard entry → Reflect on Monitor

← within same Machine - Handled by OS - Kernal not a part of CN



- CN is making data acces easy, feels like it same machine but actual client & Server are not in same machine.

Functionalities

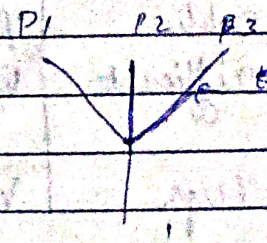
- Mandatory
  - Error Control
  - Flow Control ← Flow of Data
  - Multiplexing (MUX), Demux Demultiplexing
- Optional
  - → Encription / Decription ← Cryptography
  - → Check point - 500 file, 300 M - fail, then it should start from 300 M

using more functionalities may decrease data transfer speed.

sending data in some other form

we can't ignore

M → M1
N → M2



tells which process is having data

- Functionalities

- Ab phir se test nhi dena Pdega

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OSI Model →

TCP / IP - Model

Application
Presentation
Session
Transport
Network
Data link
Physical layer

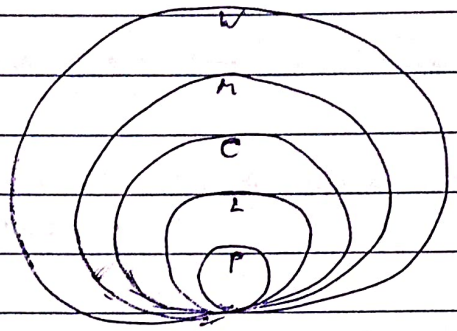
- layers

↳ the purpose of OSI model is to facilitate communication b/w different systems without requiring changes to the logic of the underlying hardware & software.

FDDI - Fibre Distributed Data Interface

CDDI - Copper

ATM - Asynchronous transfer data mode technique



	PAN	LAN	CAN	MAN	WAN
Full form	Personal Area Network	Local Area Network	Campus Area Network	Metropolitan Area Network	Wide Area Network
Technology	Bluetooth, IrDA (wireless), Zigbee, USB (wired)	Ethernet and Wi-fi, twisted pair cable	Ethernet	FDDI, CDDI ATM	telephone Leased Line, Dial-Up, Satellite link
Range	1-100 Meters (10)	Up to 2 KM	1-5 KM	5-50 KM	Above 50 KM
Transmission speed	Very High	Very High	High	Average	Low
Area	Within a Room	Within office, building	Within University, corporate office	Gov agencies use Within City like Mumbai	Within Countries
Ownership	Private	private	private	Private or Public	private or public
Maintenance	Very Easy	Easy	Moderate	Difficult	very difficult
Error Rate Cost	Very Low	Low	Moderate	High	Very High

High Security